What are Babel, esbuild and Webpack ? Explain Bundles

## Babel

=> Babel is JS compiler
=> It convert modern JS into compatible JS code
=> It helps developers use the latest JS features
=> It can transform JSX into JS
=> It support modern JS syntax
=> It improve compatability with different browser
=> It helps developers write modern JS code

## Esbuild

* esbuild is a fast JS bundler and build tool
* It is written in Go
* It ~~contains~~ Combines multiple JS files
* It supports JXS and Typescript
* It can process CSS and other assets
* It provides very fast compilation
* It helps create optimized application files

## Web pack

* webpack is a JS module bundler
* It can combines multiple ~~module~~ files into bundles
* It manages project dependencies
* It supports JS, CSS, images, and other assets
* It supports code splitting
* we can use tree shaking to remove unused code
* It optimize applications for production.

1] Website

→ You can view a website using web browser like chrome, safari or firefox by typing in its URL or following a hyperlink

→ A website is a collection of one or more related web pages identified by a common domain name & hosted on a web server.

Types of website:

Static website - content does not change automatically.

Dynamic website - content can change based on user interaction or data.

Eg: Google, Youtube, Amazon

Main components of website

Header - Top section of a website

Navigation Bar - Contains links like Home, About, Services, Contact.

Main content - Display text, images, videos etc.

Footer - Bottom section containing contact details & other information.

Technology in Website

HTML → Structure of ~~content~~ website

CSS → Design & styling

JS → Functionality & interactivity

Bootstrap → Responsive design

→ Website is a collection of related webpages hosted on web server & accessed through the Internet using web browser.

2] Webpage:

→ A webpage is a single document or page available on the Internet.

→ We can open and view a webpage using a web browser like Google chrome.

A webpage is a single page of a website that contains information such as text, images, videos, links & forms.

Eg Imagine a college website. It may contain different web pages.

3] Attributes:

→ An attributes provides additional information abou an HTML element.

→ Attributes are always written inside the openi tag

Syntax

<tagname attribute = "value">
content
</tagname>

Types of Attributes

→ Global Attributes
→ Local Attributes

Global Attributes:

Global Attributes can be used on almost every HTML element

Important Global Attributes:

Id Attribute → uniquely identify an element
class Attribute → used to group multiple element
Title Attribute → Displays tooltip information
Style Attribute → used for inline CSS.

Local Attributes:

→ work on specific element

Eg href, src, alt, target.

4) CSS layout:

CSS layout is used to arrange & position HTML elements on a webpage

1) Normal flow

Arrange elements in the default order.

Properties:

display: Block.
display: inline
display: inline-block.

Flexbox :

→ Arrange elements in a row or column

→ Properties

- display - flex
- flex-direction
- justify-content
- Align-items
- gap

Used

Navigation, bars, menus, buttons & product cards.

Grid: - Properties: display: grid, grid-template-columns, gap, grid-column, grid-row

Arrange elements in rows & columns.

3] Programming language:

A programming language is language used to communicate with computers & give instructions to perform specific tasks.

Eg C
C++
Java
Python
Javascript.

High level → Easy to understa
Eg: Python, Javascipt

Low level → closer to compute hardware
Eg: Machine language & Assembly langu

4] Javascript:

→ high level programming language used to make webpages interactive & dynamic

→ HTML creates the structure, CSS provi the design, & Javascript adds functionality

Features of Javascript:

Lightweight: It is easy to use & does not require complex setup for basic programs.

Dynamic variables can hold different types of values

Object Based: It supports objects & prototype -based Inheritance.

Asynchronous: It supports operations like API calls without blocking normal execution.

Use of Javascript:

* Form validation
* Interactive webpages
* API communication
* Animations.
* Web & mobile Application development.

6] Functions in JS.

A function is a block of reusable code designed to perform a particular task. We can execute a function whenever we need it by calling its name.

Eg

```
function greet(){
console.log("Hello world");
}
greet();
```

Types of function in JS

1. Function Declaration

A function defined using the function keyword

Eg

```
function add() {
  console.log(10+20);
}
add();
```

2] Function expression:

A function stored inside a variable

Eg

```
const add = function() {
  console.log(10+20);
};
add();
```

3] Arrow function

A shorter way of writing functions using the arrow (=>) symbol:

```
const add = () => {
  console.log(10+20);
};
add();
```

(7) Note La iruku → The fetch built in JS Interface used to make http request & commu with server.

→ It supports methods such as GET, POST, PUT, PATCH, & Delete.

The fetch () function accepts two parameters:

1) The URL to send the request to (this is a required parameter).

8] How to fetch API data using Fetch:

Eg → fetch API data from server or API
→ Returns a promise

```
fetch("https://jsonplaceholder.typicode.com/users")
  .then(response => response.json())
  .then(data => console.log(data))
  .catch(error => console.log(error));
```

Sends a request to API

Converts the response into JSON data

Successful promise

error

9] React.js is a js library used to build UI, especially single page applications.

Features:
- component-based architecture
- Reusable components
- Virtual DOM.
- JSX support
- One-way data flow
- State & prop.

10] → In this note.

| npm | npx |
| --- | --- |
| Installs & manages package | Executes package |
| used to install dependencies | used to run package commands |
| Eg: npm install | eg: npx vite |
| npm - node Package Manager | npx - node package extension. |

10] Babel, esbuild & webpack? Explain bundlers

Ans These are JS development tools used to prepare application for browser

Babel: converts modern JS and JSX into code supported by target environments.

→ Babel is JS compiler → It can transform JSX into JS.

→ It helps developers write modern JS code.

Esbuild: → Esbuild is a fast JS bundler and build tool → It is written in Go

→ It supports JSX & Typescript

webpack: → webpack is a JS module bundlers

→ It supports JS, CSS, images & other assets.

→ It manages project dependencies.